Airbnb Data Analytics using Redis

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Problem & Motivation

- Airbnb generates large volumes of listing, pricing, and availability data.
- Efficient analysis is required to support business-oriented insights.
- The course requires the use of Redis to explore in-memory analytics.
- Objective:
    a. Build an automated pipeline to process Airbnb data and answer analytical queries.
    b. Transform raw Airbnb data into actionable BI insights.

Dataset Description

Data Source	City Selected	Datasets Used
InsideAirbnb (public open dataset)	**New York City,** **USA**	**Listings dataset** (property details, prices, ratings, location) **Calendar dataset** (availability information) **Reviews dataset** (Reviews by Guests)

- New York City was selected due to its large and diverse Airbnb market.

Data Preprocessing & Cleaning

1. Initial exploration of Airbnb listings and calendar datasets
2. Removal of irrelevant and redundant attributes
3. Selection of key attributes (e.g., listing ID, price, rating, room type, neighbourhood, location)
4. Handling missing values:
   - Missing prices filled using average daily price from the calendar dataset.
   - Missing textual fields replaced with empty strings to ensure safe storage.
5. Data type normalization and preparation for efficient Redis storage

Pipeline Overview

- Automated end-to-end data processing pipeline
- Data ingestion from InsideAirbnb datasets
- Data preprocessing and feature selection
- In-memory storage and indexing using Redis
- Analytical queries and visualizations generated from Redis

InsideAirbnb Data

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Data Preprocessing

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Redis Data Store

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Queries & Visualizations

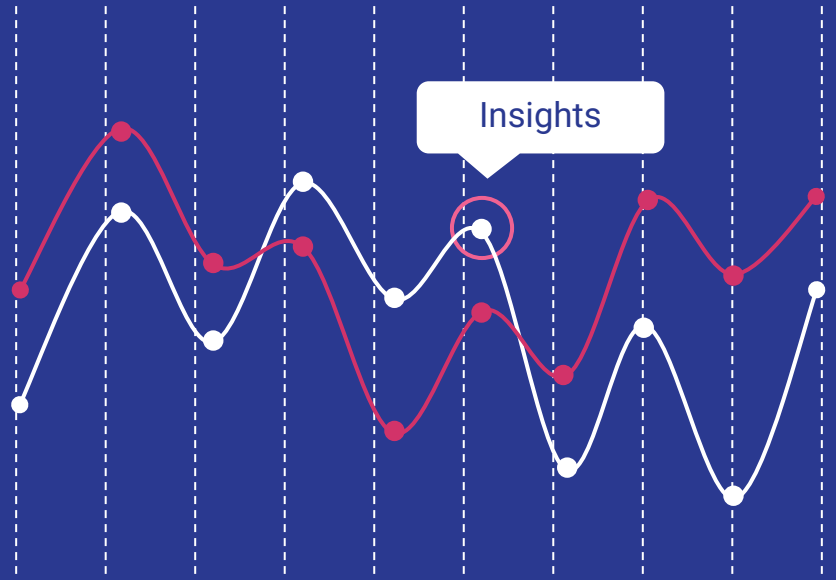
Redis Data Model

1. **Hashes**
   - Enable fast filtering and aggregation, Example: listing:<id> → name, price, rating, room type, neighbourhood
2. **Sets**
   - Group listings by city and neighbourhood, Enable fast filtering and aggregation
3. **Sorted Sets**
   - Store listings ranked by rating, Support efficient top-N queries
4. **GEO Index (extensible)**
   - Store latitude and longitude of listings, Enables future spatial and proximity queries

Queries Implemented

1. Ranking of top Airbnb listings based on rating and price
2. Neighbourhood-level market analysis using aggregated KPIs
3. Availability and pricing analysis using calendar data
4. Occupancy analysis using availability percentage
5. Amenities frequency analysis
6. Amenities analysis for identifying features of highly rated listings

Key Outputs, Visualizations & Insights



Query 1: Top Listings Ranking

Key Insights:

- Best-value listings are not necessarily expensive.
- High rating + moderate price defines competitive listings.

Top 10 Highest Rated Listings in NYC (Rating DESC, Price ASC)

Listing ID	Rating	Price ($)	Room Type	Neighbourhood	Name (Shortened)
995268785620684770	5.00	41.0	Private room	Bedford-Stuyvesant	Bright bedroom with high ceiling in Beds...
988787058900567282	5.00	41.0	Private room	Bedford-Stuyvesant	Luxury Duplex Apt, HDTV, laundry, terrac...
991301559931083002	5.00	44.0	Private room	Harlem	Beautiful Private room
984866964862347137	5.00	44.0	Private room	Bay Ridge	stylish (balcony)
992033931066536338	5.00	45.0	Private room	Harlem	Affordable private room with shared bath...
995095494292465041	5.00	48.0	Private room	Washington Heights	Comfy Room in Manhattan close to Subway
984670477924074027	5.00	49.0	Private room	Bedford-Stuyvesant	Comfy Bedroom
992285225039563541	5.00	54.0	Private room	Bedford-Stuyvesant	Tuk Ahoy - Sunset Suite (3B)
997093934680613043	5.00	57.0	Private room	Flushing	Lovely Exuberant Home 3A
995686208234684657	5.00	58.0	Private room	Bedford-Stuyvesant	bedstuy is fly(1)

Top 10 Listings Sorted by: Rating (DESC), Price (DESC)

Listing ID	Rating	Price ($)	Room Type	Neighbourhood	Name
9879796	5.00	629.00	Entire home/apt	Upper East Side	Modern 2 Bedroom by Memorial Sloan Kette...
990798417345656604	5.00	614.00	Entire home/apt	Midtown	Beautiful Central Park studio
998088361998367565	5.00	597.00	Entire home/apt	Chinatown	Luxury Penthouse! 2 Bed / 2 Bath + Priva...
986263425862656510	5.00	588.00	Entire home/apt	Hell's Kitchen	Gorgeous space near Hells Kitchen and Mi...
984667555924799835	5.00	526.00	Entire home/apt	West Village	Designer West Village Pool House
994794484302782803	5.00	474.00	Entire home/apt	Hell's Kitchen	Blueground | Midtown W, gym & w/d, nr Hu...
997432199888717593	5.00	419.00	Entire home/apt	Sunset Park	The Greenwood Gem - 3BR
994791843427497764	5.00	407.00	Entire home/apt	Upper West Side	Blueground | UWS, gym, close to central ...
995683781967427706	5.00	391.00	Entire home/apt	Sunset Park	Greenwood Heights Delight - 3BR
986816875695859005	5.00	390.00	Private room	Midtown	The Manhattan Club

Query 2:
Neighbourhood-level BI Analysis

Key Insights:

- High supply does not imply high ratings.
- Premium areas often show lower satisfaction.
- Affordable neighbourhoods achieve competitive ratings.

Top 10 Neighbourhoods

	neighbourhood_cleansed	total_listings	avg_price	avg_rating
13	Bedford-Stuyvesant	2612	132.180513	3.289548
217	Williamsburg	2049	470.362128	3.457706
131	Midtown	1993	2573.406422	2.584124
97	Harlem	1685	155.001484	3.547181
98	Hell's Kitchen	1496	1391.211230	2.982981
205	Upper West Side	1458	1238.795610	2.655501
28	Bushwick	1443	105.699931	3.107277
204	Upper East Side	1399	1051.037884	3.106605
53	Crown Heights	1133	168.517211	3.331253
66	East Village	959	233.087591	3.334025

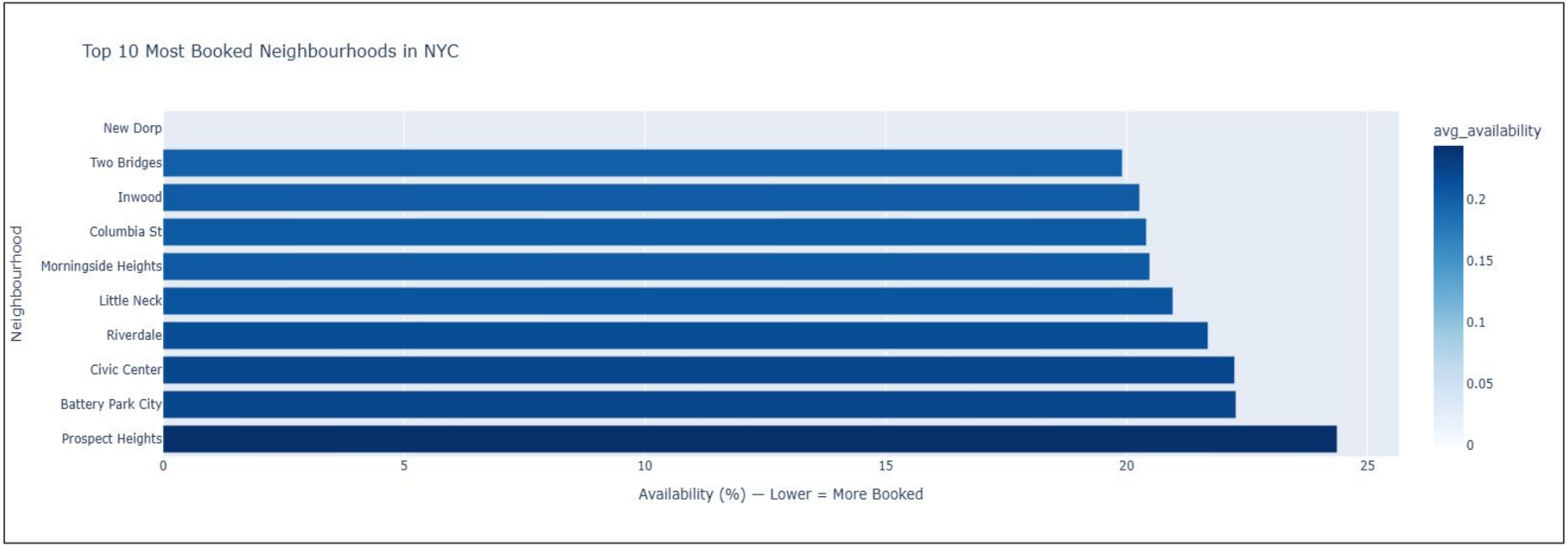
Query 3:
Availability & Pricing Analysis

Key Insights:

- Low availability → high demand
- Central neighbourhoods show higher booking intensity.

Availability by Neighbourhood

	neighbourhood_cleansed	avg_availability	availability_pct
145	New Dorp	0.000000	0.00
201	Two Bridges	0.199087	19.91
107	Inwood	0.202690	20.27
46	Columbia St	0.204110	20.41
134	Morningside Heights	0.204832	20.48
120	Little Neck	0.209589	20.96
170	Riverdale	0.216895	21.69
38	Civic Center	0.222356	22.24
6	Battery Park City	0.222706	22.27
162	Prospect Heights	0.243682	24.37


Top 10 Most Booked Neighbourhoods in NYC
New Dorp
Two Bridges
Inwood
Columbia St
Morningside Heights
Little Neck
Riverdale
Civic Center
Battery Park City
Prospect Heights
Neighbourhood
0
5
10
15
20
25
Availability (%) — Lower = More Booked
avg_availability
0.2
0.15
0.1
0.05
0

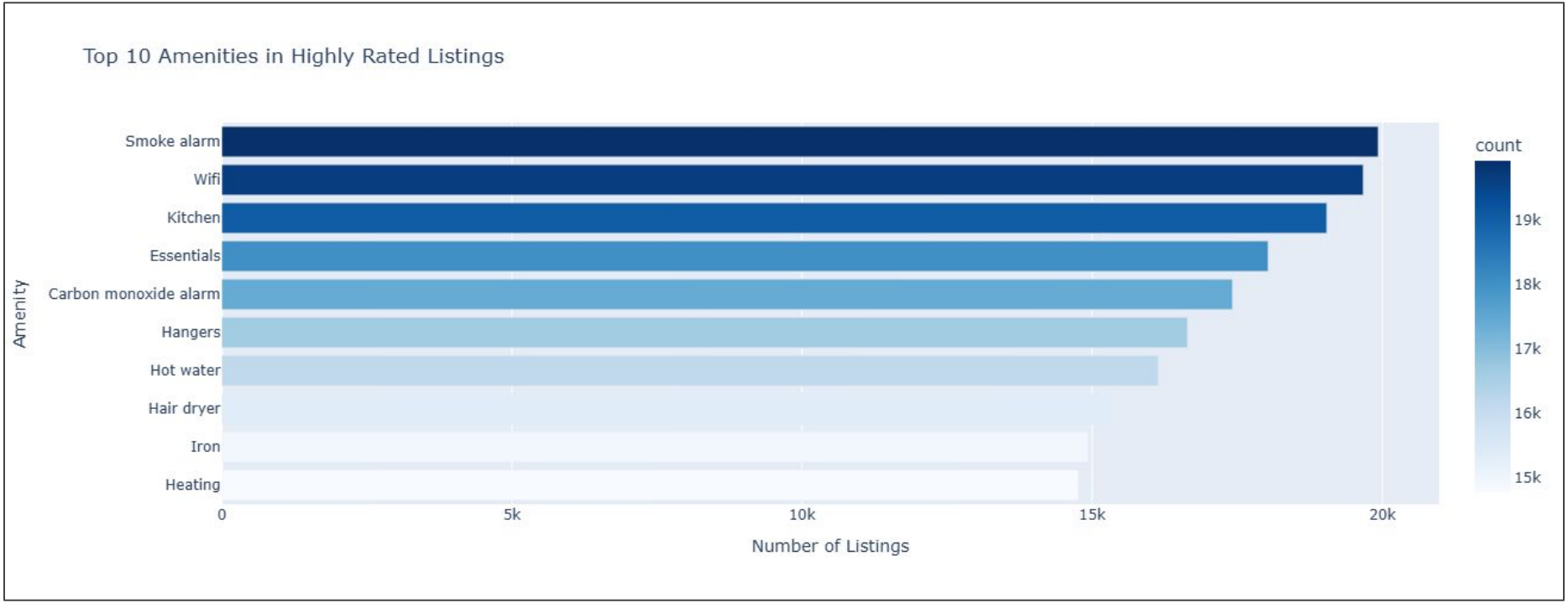
Query 4:
Amenities Analysis

Key Insights:

- Basic amenities like Wifi and Kitchen are expected.
- Comfort-oriented amenities correlate with higher ratings.

Amenity Counts

	amenity	count
12	Smoke alarm	19930
14	Wifi	19671
28	Kitchen	19044
36	Essentials	18032
15	Carbon monoxide alarm	17419
22	Hangers	16643
21	Hot water	16141
29	Hair dryer	15349
4	Iron	14931
19	Heating	14765


Top 10 Amenities in Highly Rated Listings
Smoke alarm
Wifi
Kitchen
Essentials
Carbon monoxide alarm
Hangers
Hot water
Hair dryer
Iron
Heating
Amenity
0
5k
10k
15k
20k
Number of Listings
count
19k
18k
17k
16k
15k

Stakeholders & Value

Stakeholders

- Travelers
- Hosts
- Airbnb Platform

Business Value

- Better travel decisions
- Pricing and amenity optimization
- Market demand understanding

Conclusion

- Overall, the combination of Redis-based data storage and business intelligence analysis enables fast querying and meaningful insight extraction from large Airbnb datasets.
- The project highlights how availability, pricing, and amenities can be jointly analyzed to understand market dynamics and customer preferences in a major urban market such as New York City.

Thank you :)